Firewatch: catch the grass fire in the 27 minutes nobody knows.

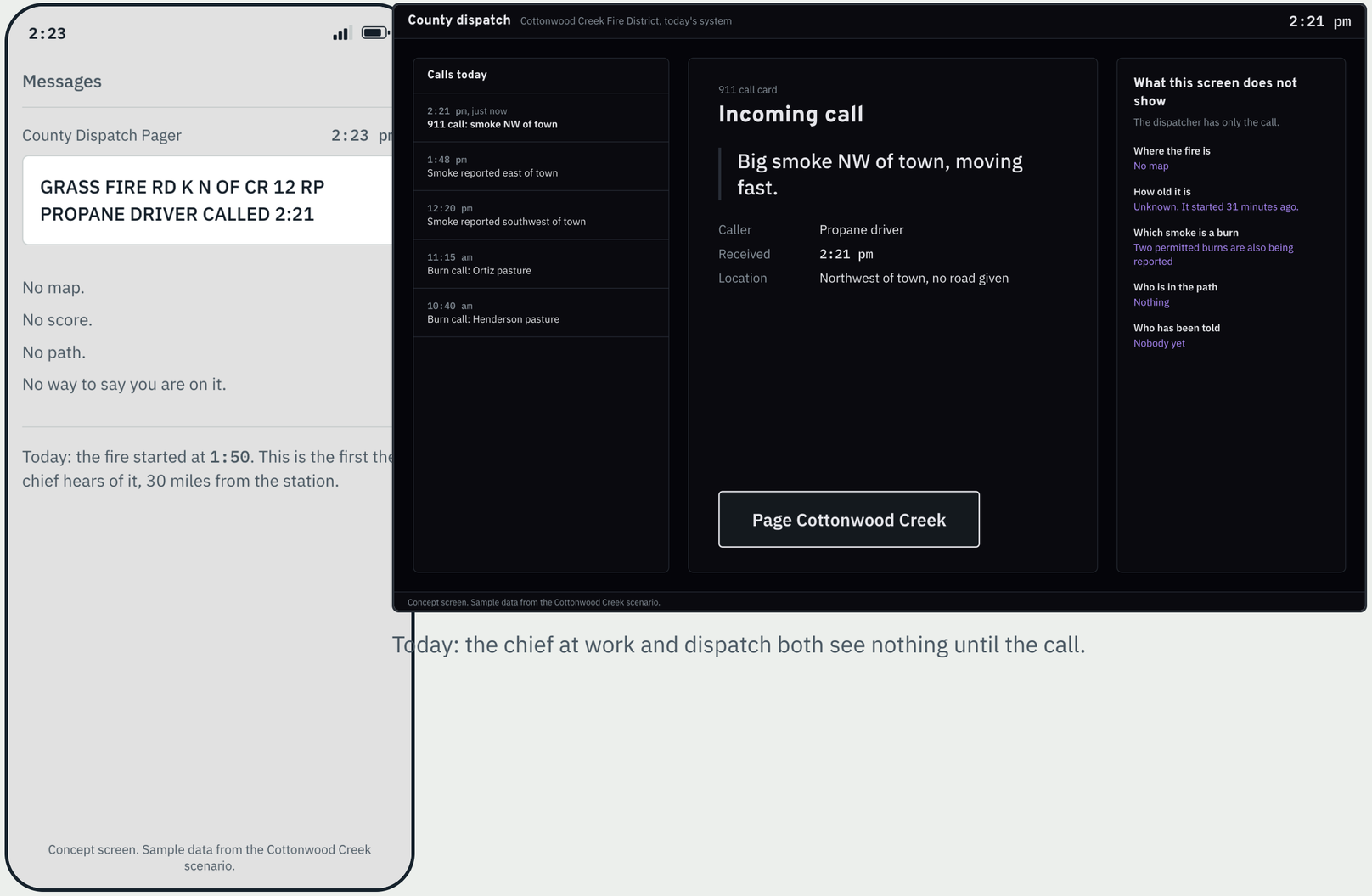
Team Watch Dogs

Firewatch 2026



Nobody knew for 27 minutes, and the first truck arrived 62 minutes after the line fell.

1:50	Line falls into grass. Nobody knows.
2:17	A driver sees smoke. No signal.
2:21	911 call. District paged.
2:34	Brush 1 rolls.
2:52	Brush 1 on scene. Head 2 to 3 miles out.
3:10	Mutual aid requested.



Today: the chief at work and dispatch both see nothing until the call.

Source: Cottonwood Creek scenario (sample data)



Three barriers stand between the fire and the first truck, and each shows up in this afternoon.

Time

The fire burns 27 minutes before anyone knows. The chief is 30 miles away, and the 911 caller had no signal for 3 miles.

First page 2:21, 31 minutes after ignition.

False alarms

Smoke is normal in March. Two permitted pasture burns are on file with dispatch, so a smoke alert alone would page the district for nothing.

Two legal burns the same afternoon.

Cost at rural scale

A volunteer district with a mill levy cannot buy cameras or a sensor field. Camera detection runs about $50,000 per station per year.

Source: Pano AI pricing, TechCrunch 2023

62-minute clock



Firewatch targets a page by 2:05 and a truck on scene by 2:34, 18 minutes sooner than today.

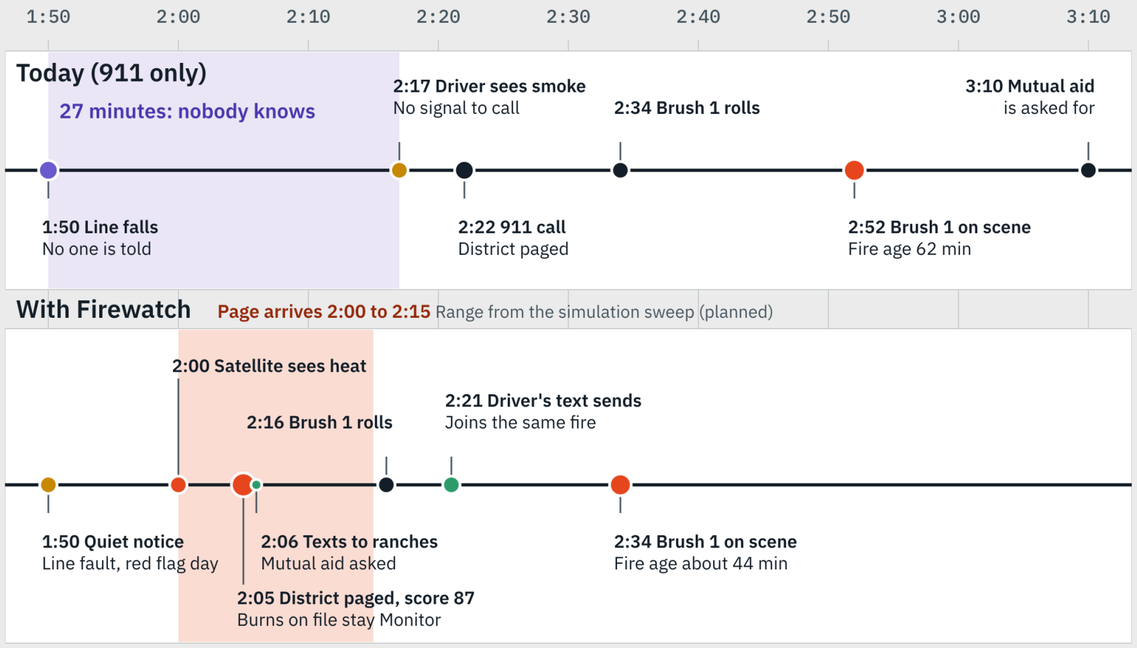
Today: 911 at 2:21, on scene 2:52.

Firewatch: paged about 2:05, on scene about 2:34.

These are targets from the planned simulation. They are not results yet.

Tuesday afternoon, replayed: 911 only vs Firewatch

Same fire, same weather, same crews. Only the first alert changes.



Fire age when Brush 1 arrives

Today (911 only)
62 min

With Firewatch
about 44 min

About 18 minutes younger, and the crew is sent to the east flank on CR 12.

Targets, not measured results

These numbers come from the planned simulation. The simulation is not built yet. We will report the range, not one lucky run.

Concept screen. Sample data from the Cottonwood Creek scenario.

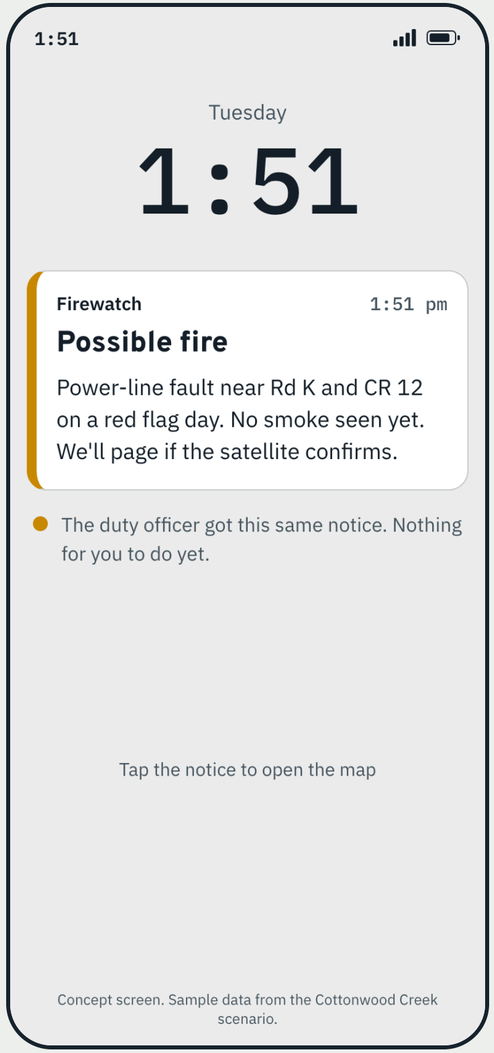
62-minute clock



At 1:51 a line fault on a red flag day sends a quiet "Possible fire" notice, and nothing more.

The chief and duty officer see the fault, the weather and the wait for the satellite. No page goes out yet.

A fault alone is only a watch, so windy days do not wear out the alert.



Source: Cottonwood Creek scenario (sample data)

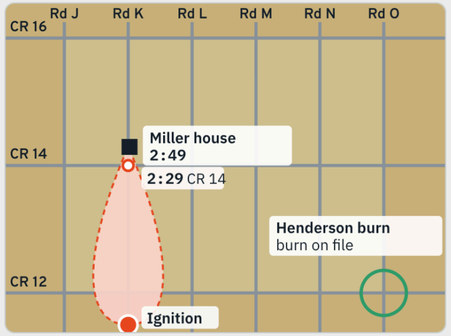


At 2:05 the satellite confirms it and the district is paged. The two burns stay quiet.

2:05

Grass fire. Page the district.

87 out of 100. Sure enough to page.



Reaches CR 14 by 2:29, Miller house by 2:49

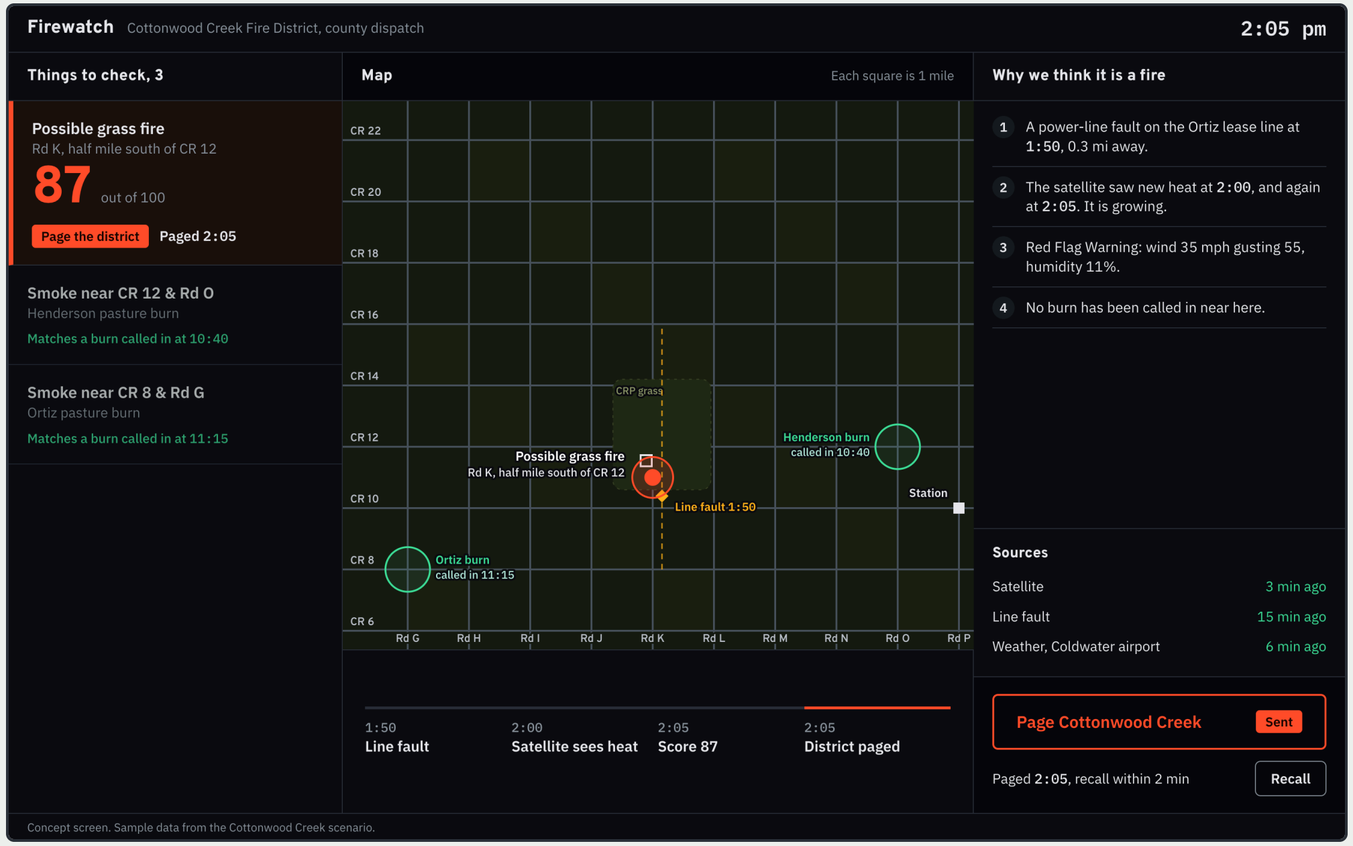
1. Power-line fault on the Ortiz lease line, 0.3 mi away	15 min ago
2. Satellite saw new heat at **2:00** and **2:05**, and it is growing	3 min ago
3. Red Flag Warning: wind 35 mph gusting 55, humidity 11%	6 min ago
4. No burn has been called in near here	Dispatch, now

I'm on it

Request mutual aid

Not a fire

Concept screen. Sample data from the Cottonwood Creek scenario.



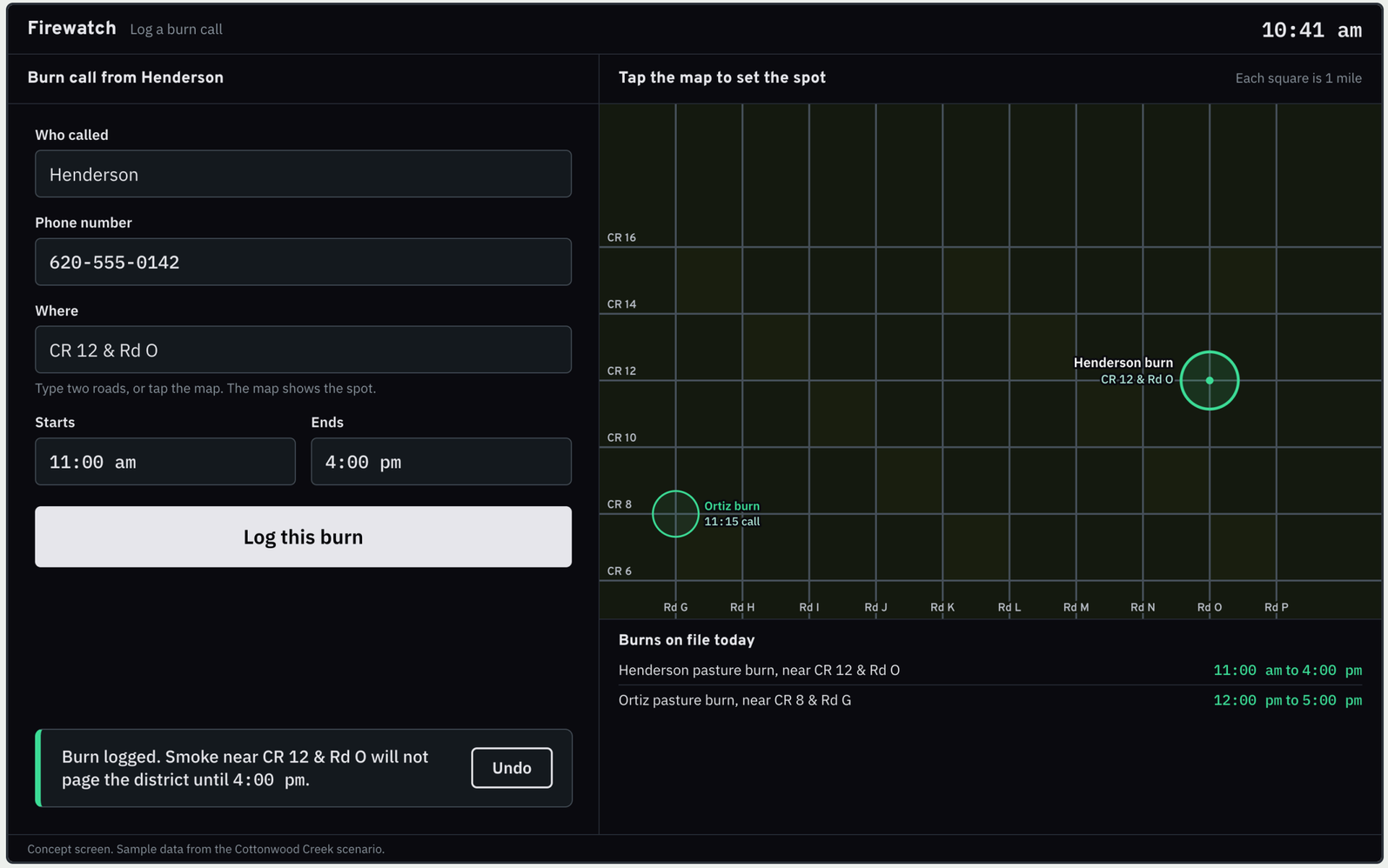
The dispatcher sees the same ranked queue, with the reasons beside each score.

Score 87. Page the district.

Henderson and Ortiz burns stay "Monitor".



Dispatch already takes burn calls, so Firewatch reads them to tell a real fire from a permitted burn.

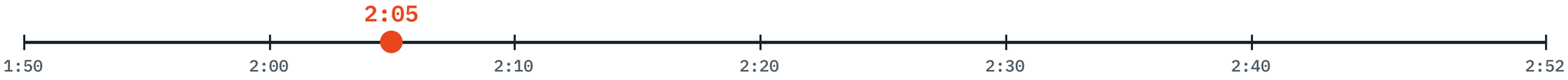


The Henderson burn was called in at 10:40. The Ortiz burn at 11:15.

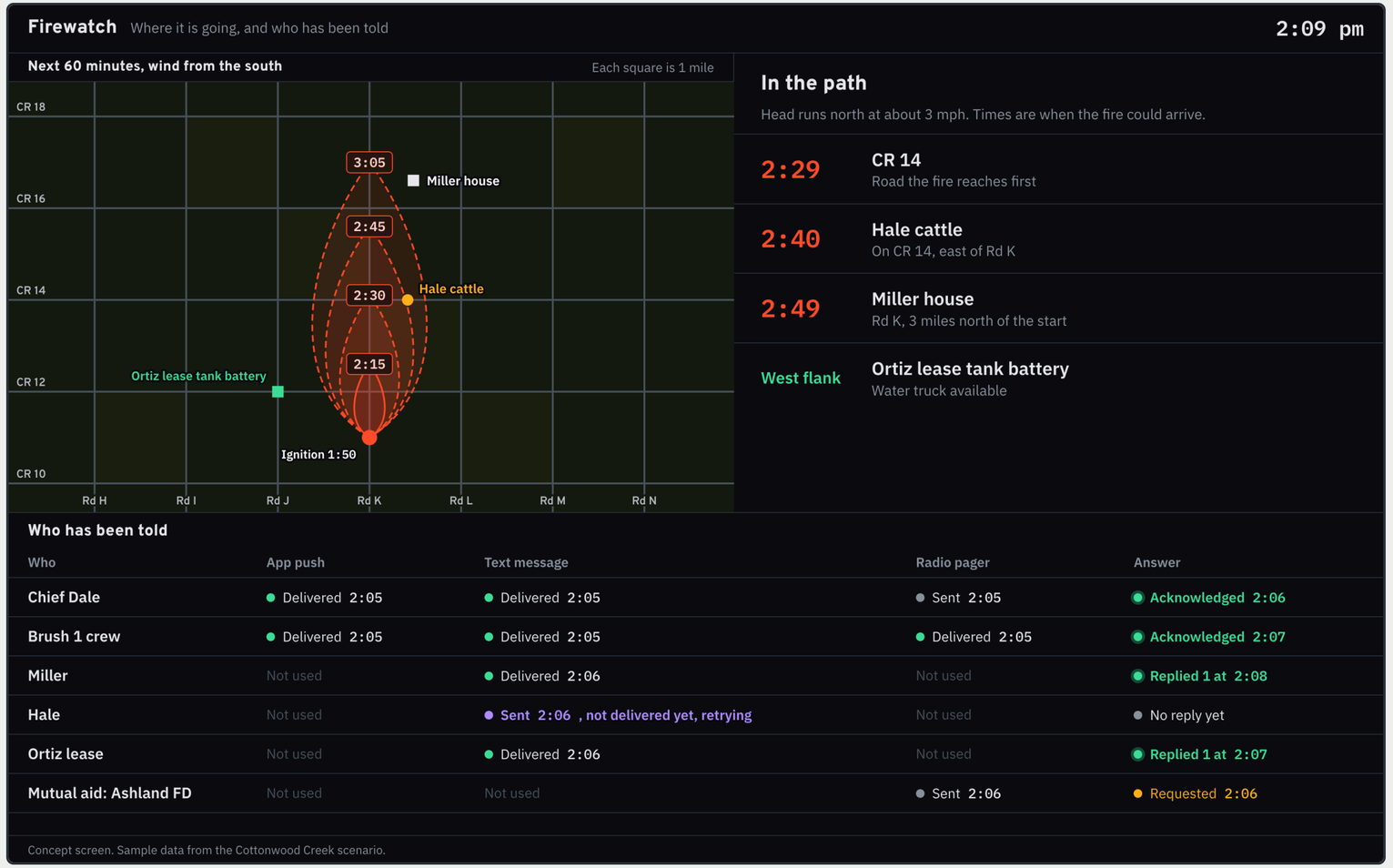
A hotspot near a logged burn scores down and says why. A burn missing from the log can still alert, and one tap dismisses it.

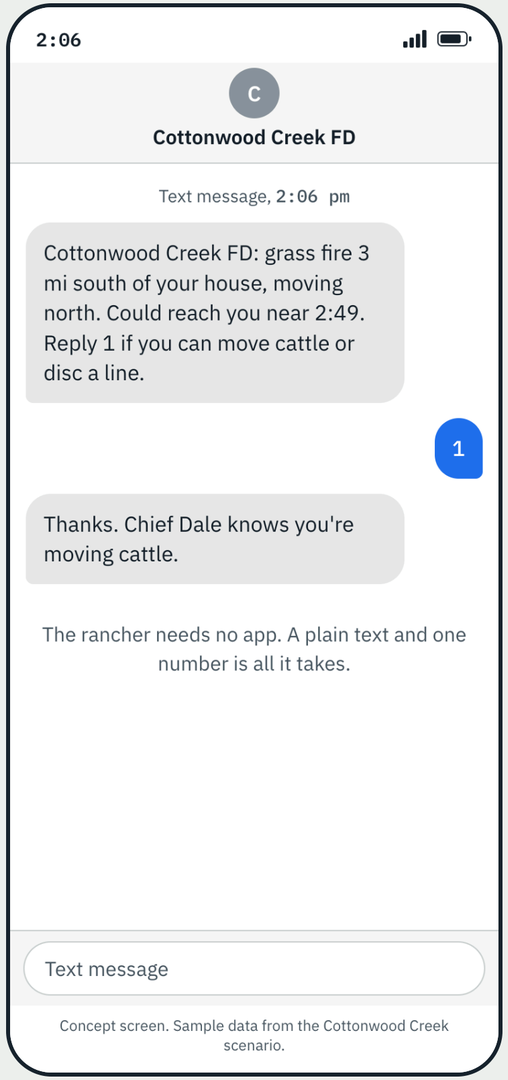
Source: Barber County burn resolution as reported by KCN. We will confirm the wording with the county clerk.

62-minute clock



Respond means more than the station: the ranchers in the path get a text too.

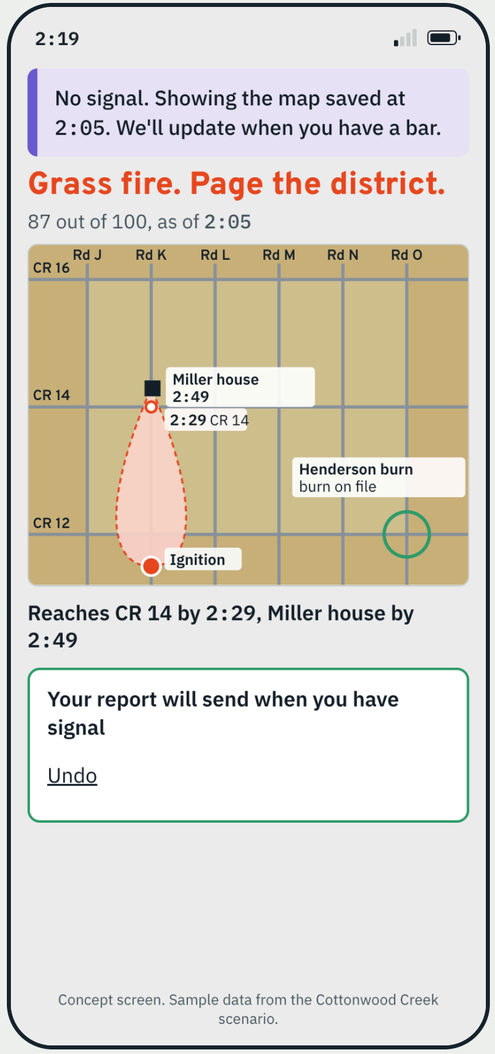




The 60-minute forecast shows the Miller ranch reached near 2:49. Miller, Hale and the Ortiz lease get a text at 2:06 and can move cattle or start a disc line.



Built for one bar of signal: short texts, a radio pager that needs no cell, and a map stored on the phone.



Built for one bar of signal

Every alert goes out every way at once. The ones lower down need less from the phone.

The delivery ladder

All routes fire together. If one is not acknowledged, the next is tried.

1	**App push**	A short alert with a link to the map	Needs data
2	**Text message**	Under 160 characters. Held until a bar comes back.	Needs one bar
3	**Radio pager, through county dispatch**	One line, sent by email-to-page. The backbone.	Needs no cell at all
4	**LoRa mesh radio**	Short text from the station to trucks and ranches. Optional.	Needs no cell at all
5	**Satellite text**	Later. Next step, price not confirmed.	Needs no cell at all

How big each message is

Our estimates, to be measured

In the field, over one bar

Text alert	under 160 characters
Full alert	about 3 KB
Spread shape	about 8 KB

Ahead of time, over station WiFi, once

Map cached ahead of time	about 20 MB

About 2,500 times the spread shape, which is why it is sent once, not on the day. The bars above use one scale.

The heavy work happens where the bandwidth is. The field only has to receive small messages.

Concept screen. Sample data from the Cottonwood Creek scenario.

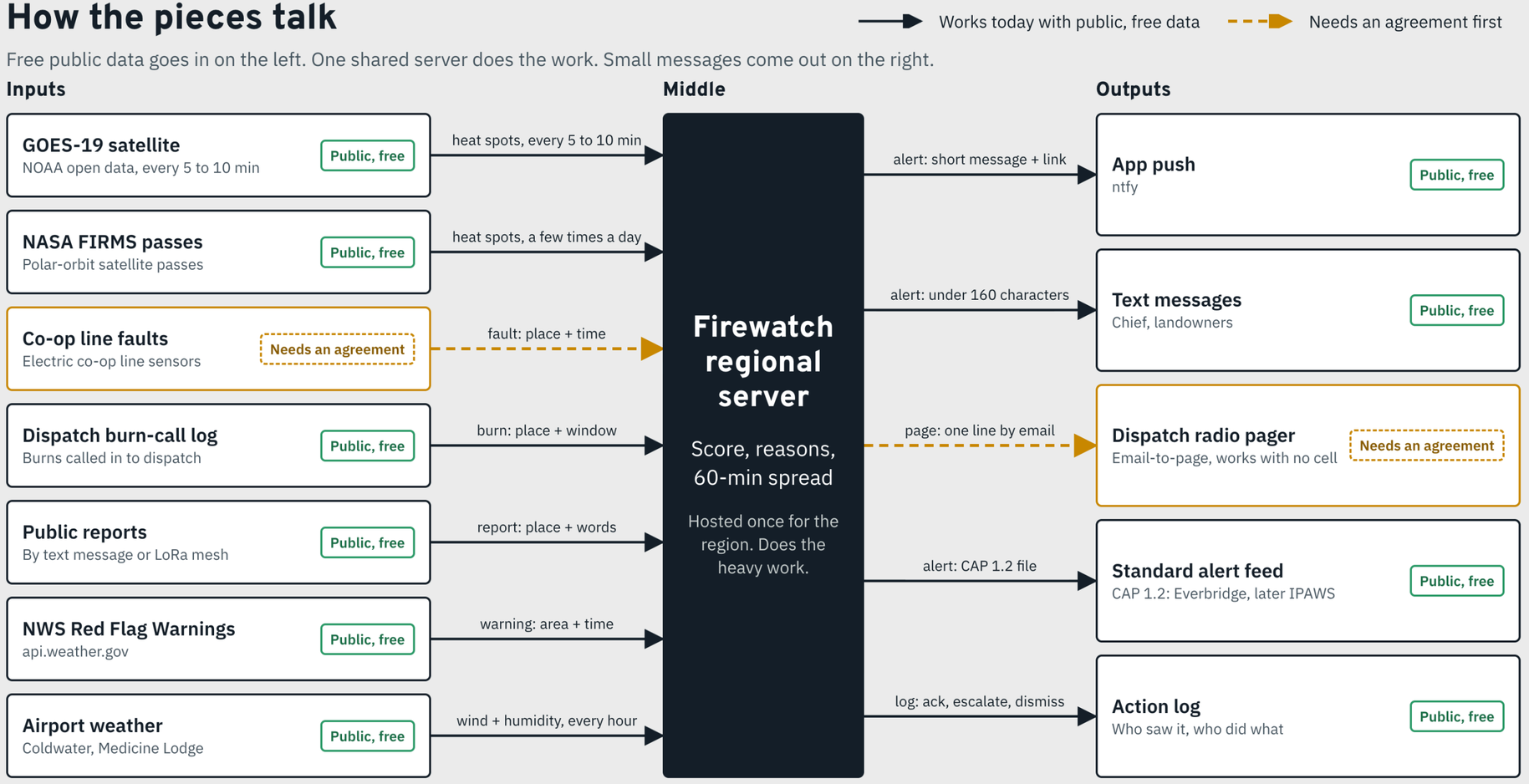
62-minute clock



The pieces connect through one shared server, and only two links need an agreement.

How the pieces talk

Free public data goes in on the left. One shared server does the work. Small messages come out on the right.



Concept screen. Sample data from the Cottonwood Creek scenario.

Solid line

Works today with free, public data.

Dashed line

Needs an agreement first: the co-op line faults and the dispatch radio pager.

Source: NOAA GOES-19 fire product and NASA FIRMS (Earthdata, ARSET 2025)

62-minute clock



The core is built. We add inputs, burn handling, delivery and a phone screen.

Detect

FIRMS satellite passes

Built

GOES-19 satellite, direct

To build

Co-op line-fault input

To build

Public reports by text

To build

Confirm

Score with reasons

Built

Dispatch burn-call log

To build

Red flag and corroboration

To build

Data age for every source

Partly

Respond

60-minute spread forecast

Built

Who is in the path

Partly

Push, text and pager alerts

To build

Standard alert export (CAP)

To build

Field

Phone screens for the chief

To build

Offline district map

To build

One-tap replies

To build

Mesh radio, optional

To build

Built means it runs in the current app. Partly means a piece exists, such as two source chips or a nearest-asset list.

62-minute clock



When a feed fails, the screen says so and 911 stays the main path.

What it does when something fails

A missing feed shows up as a gap, never as "no fire".

Feed health, as dispatch sees it

Example of a bad moment, not the whole afternoon.

Satellite is late.
No new data is not the same as no fire. 911 is still your main alarm.

Satellite	Late, last scan 22 min ago
Co-op line faults	Offline since 1:20
Weather	OK, Coldwater airport 6 min ago
Burn log	OK, 2 burns today
Text messages	OK
Radio pager	OK

Five things that can go wrong

What you see, and what keeps working.

The satellite is late or down

What you see
A banner with the age of the data, and a gap on the map. Never "no fire".

What still works
911 is still your main alarm. Line faults, weather and reports still count.

The co-op feed is down

What you see
Line fault input greyed out

What still works
The satellite and public reports still score.

The burn log misses an entry

What you see
The alert says "no burn logged here"

What still works
Dispatch dismisses it in one tap. Each dismissal is logged.

No cell signal in the field

What you see
Short texts, under 160 characters

What still works
Offline map, the radio pager, and LoRa mesh.

The score is wrong

What you see
A score with its reasons

What still works
A person decides. Every override is logged.

Concept screen. Sample data from the Cottonwood Creek scenario.

If this fails	This still works
Satellite late or down	Banner shows data age. A gap is never shown as "no fire".
Co-op feed down	Fault input greys out. Satellite and reports still score.
Burn missing from log	It can alert, says "no burn logged here", and one tap dismisses it.
No cell in the field	Short texts, offline map, pager, optional mesh.
Score is wrong	A human decides and every override is logged.

62-minute clock

1:50 2:00 2:10 2:20 2:30 2:40 2:52

We name the risks, rate them, and keep a fallback for each.

Risk	Likelihood	Impact	How we reduce it
The co-op line has no fault telemetry (old oil-lease taps may have only fuses)	Medium	High	Satellite alone still pages about 2:05 to 2:15. Ask the co-op to share what it can see.
GOES misses or is slow on a small early fire (to be measured)	Medium	High	Log scan, file and ingest times. Test Starbuck. Keep FIRMS and reports as backups.
Fault-only alerts on windy days cause alert fatigue	High	Medium	A fault alone is a quiet watch. Only fault plus satellite or a report pages.
The burn log is incomplete or late	Medium	Medium	Reason text says "no burn logged here". One-tap dismiss, logged. Keep the old rules as backup.
SMS carrier registration or filtering delays texts	Medium	High	Send push, text and pager at once. The pager needs no cell.
Dispatch or the co-op will not sign a data agreement	Medium	High	Start with free public data. Offer a one-page pilot. Keep manual entry.
Landowner phone numbers are private	Medium	Medium	Use the county Everbridge opt-in list. Store numbers only for the district.
The regional server goes down	Low	High	Pager and 911 are untouched. A status banner shows the outage. Redeploy from one script.
People over-trust the score	Medium	High	Show the reasons and data age. A human decides. Log every override.

Likelihood and impact are our own estimates, to be checked in the test.



The district pays $0 for software, while camera detection costs $50,000 per station per year.

$0

for the district. Free public data, hosted once for the region. Hosting is about $10 to 25 a month for the state (estimate, to confirm).

$50K

per station per year for camera detection. "The typical pricing is $50,000 per station per year per customer."

Who pays
Region or state hosts the software.
Co-ops pay for their line sensors.
Optional hardware: Kansas Forest Service VFA (matching funds) or USDA CWDG (10 to 25% match).

Source: Pano AI, TechCrunch 2023

Time to stand up: about a day (our estimate) for district boundary, roads, contacts, pager email and burn entry. The co-op feed takes weeks (our estimate) of agreements.

Why not new sensors: a 2026 DHS audit found N5 wildfire sensors "did not consistently detect fires or alert state and local partners", and wind hurt them.

Source: Kansas Forest Service VFA; USFS CWDG; DHS Inspector General OIG-26-05, April 2026



We will show the evidence, then pilot with the co-op and hand confirmed fires to responders.

Evidence we will show

Historical test: first satellite look vs. the Starbuck ignition at **11:51** AM CST (NWS).

Measured GOES delay, from our own log.

A simulation sweep that reports a range of minutes saved.

The National Weather Service reports 83 percent of satellite notifications arrived before local 911 calls.

Next steps

Pilot with an electric co-op.

Add a Pyronear tower camera.

FireSat early adopters get operational data from Q4 2026.

Hand confirmed events to autonomous responders, such as those EMBERPOINT is "targeting demonstrations in 2026" for.

Source: NWS Amarillo (Starbuck); Wildfire Today (83%); HeliOps (FireSat); Lockheed Martin EMBERPOINT release, Jan 26 2026

